

Two factory Cummins CM551 / Dodge 24v VP44 calibrations

Same ECM part number and ROM date. Different silk-screen codes, different cal dates, different WOT fuel. Timing tables match. This is a decoded compare of two boxes the owner already had — not a flash image and not a write recipe.

ECM A

J90269.06

cal 062800 · 6BTA 5.9 Li

ECM B

J90268.04

cal 102198 · ISB 235

P/N 03942336

ROM 091197

ReadByNTN 0x48

ITNs 667 / 0

Z-maps decoded 29

Chr0000 KennPar (ReadByNTN)

CAN 250k 29-bit PGN EF00 cmd 0x48. Packed ITN dump, not ROM. VIN and engine serial are not published.

THE TWO BOXES

Identity, hours, miles — no VIN / ESN

ECM A · J90269.06

Plate	Cummins 6BTA 5.9, FP98455, Chrysler T-300
P/N / ROM	03942336 / 091197
Cal date	062800
Engine hours	13130:48:02
Key-on	~13724
ECM VSS miles	81

ECM B · J90268.04

Plate	ISB 235, 98453, Chrysler automotive
P/N / ROM	03942336 / 091197
Cal date	102198
Engine hours	10196:32:59
Key-on	~10240
ECM VSS miles	287919

Hours, key-on, and VSS are *runtime* counters — they differ because the boxes lived different lives. A's ~81 ECM VSS miles with 13k hours is consistent with a module that ran a long time without a speed input the ECM trusted, or that did not spend its whole life in the truck that holds it now. Those numbers are not the calibration delta.

Decode: $\text{physical} = \text{raw} \times \text{scale} + \text{add}$. RPM $\times 0.125$. Fuel Z on 5DFL $\times 0.0679348$, add -800 . Timing Z $\times 0.1171875$, add -60 . Axis prefix: first u16 BE is byte length of points. Z is Y rows $\times$ X columns. Catalog comments that say 17 $\times$ 4 / 17 $\times$ 9 are wrong on these boxes — live prefixes are 4 $\times$ 18 fuel and 11 $\times$ 18 timing.

What is the same — and the three that are not

DIFFER · 3

5DFL00ZA 4×18

5DFL01ZA 4×18

AFFLLMZA 14×21

Complete fuel request (5DFL) plus truncated AFC. Timing grids match.

DECODED SAME · 12

4DTA00ZA 11×18

4DTA01ZA 11×18

ATFLLMZA 4×18

BSTATBZA 6×6

CBDCAT 2×2

CBDSCT 2×2

CPCS02YA 3×13

CPDSZA 7×13

FLCDICZA 5×9

FLFLTBZA 15×15

FLTATBZA 12×12

IHPHTMZA 4×7

NAK / EMPTY · 14

AABSPRZA 0×0

CAICTAZA 0×13

CKEXFLZA 0×0

CKIIFLZA 0×0

HG00ZA 0×0

HG01ZA 0×0

IHHTDTB 0×12

IHLOTDTB 0×12

IHPHTZ1A 0×0

IHPRTMZA 12×12

MTTQMPZA 0×0

TCSTESZA 0×0

WSTATB 0×10

WSTAZA 0×0

Chr0000 listed them; this Dodge cal NAKed ReadByNTN.

Timing 4DTA00ZA / 4DTA01ZA (11×18) are **the same** A vs B. Conversion FLFLTBZA (15×15) is **the same**. Altitude derate ATFLLMZA is the same (flat 400). The silk-screen codes are not “the same tune with a sticker change.”

Scalars: 2172 decoded, 103 differ — many are runtime (hours, key-on, VSS, faults).

4DTA00ZA raw blob first differs at byte 480 of 512, past the 11×18 table. 5DFL* first differs at byte 114, inside the table.

WOT fuel vs RPM — A is richer mid/high

+22.691

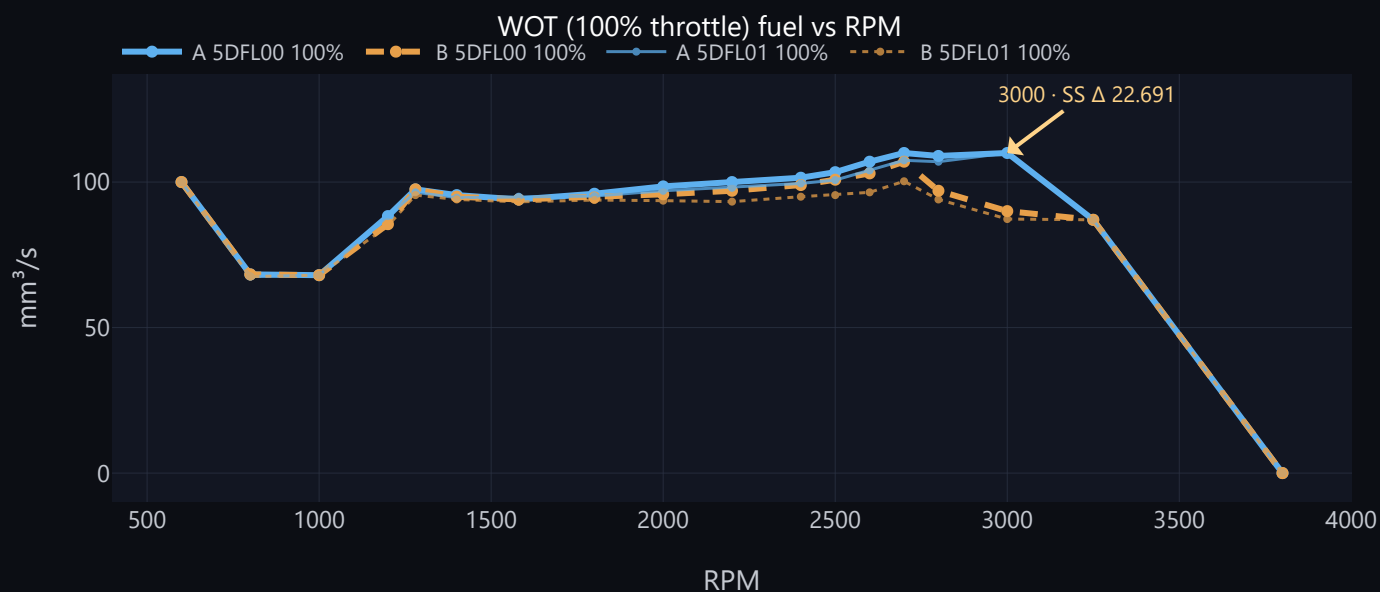
mm³/s at 3000 RPM · 5DFL01 100% (steady) · A 109.987 vs B 87.296

+19.973

mm³/s at 3000 RPM · 5DFL00 100% (transient) · A 109.987 vs B 90.014

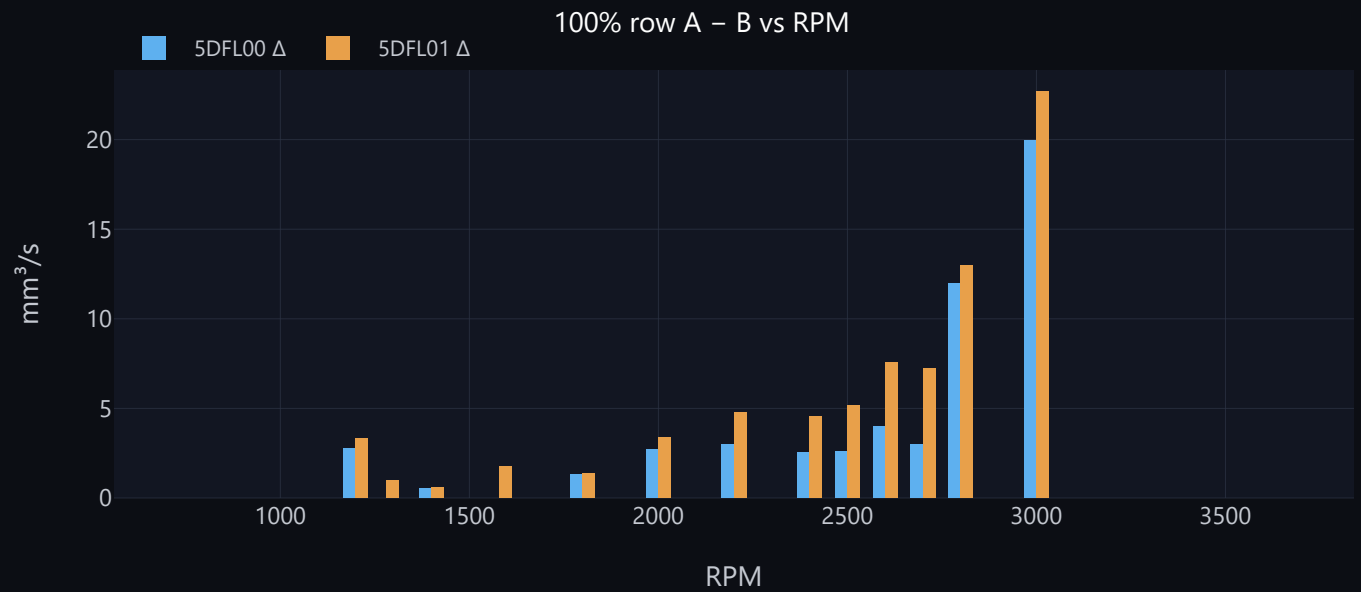
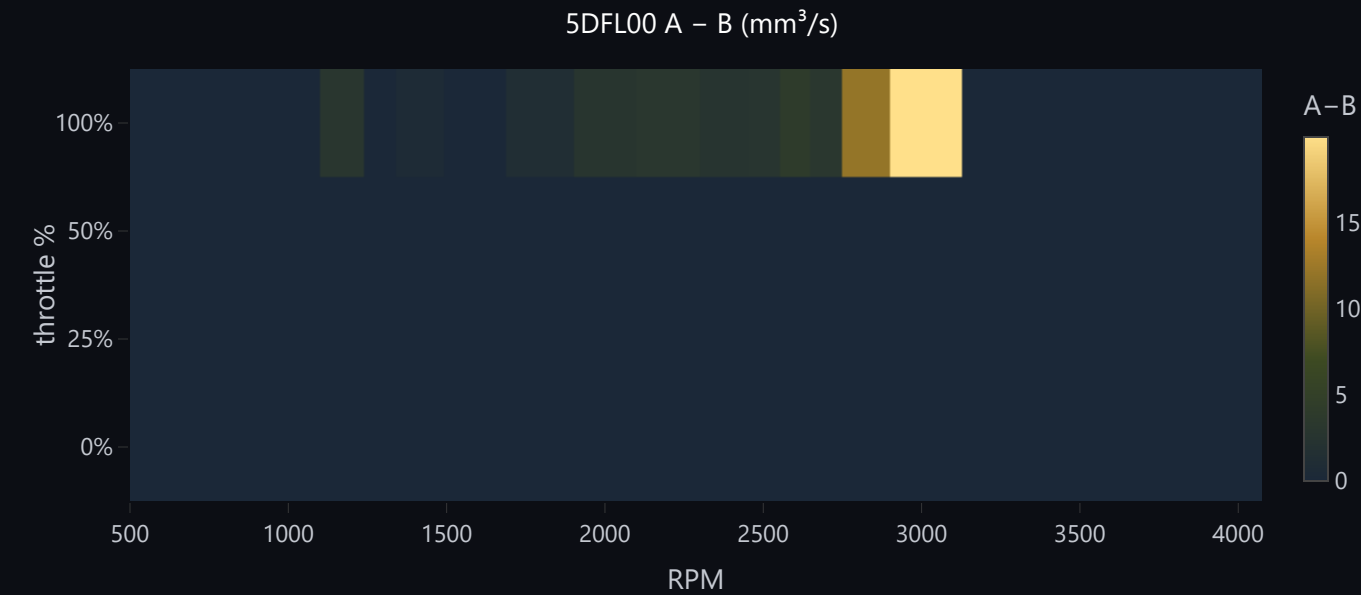
2800 RPM gaps

Transient +11.957 · steady +12.976 mm³/s. Part-load rows (0 / 25 / 50%) match. 3800 column is 0 on both (unused breakpoint).



Verified from tune_A_vs_B.json. Earlier shop note of ~+20–23 mm³/s at 3000 RPM 100% is the pair above (19.973 transient / 22.691 steady), not a third table.

A – B heatmap and the 100% tables



5DFL00ZA 100% row (MM3S)

RPM	A	B	A – B
600	100	100	0
800	68.275	68.275	0
1000	68.003	68.003	0
1200	88.315	85.53	2.785
1280	97.487	97.487	0
1400	95.517	94.973	0.544
1580	94.022	94.022	0
1800	95.992	94.701	1.291
2000	98.506	95.788	2.718
2200	100	97.011	2.989
2400	101.495	98.981	2.514
2500	103.397	100.815	2.582
2600	106.998	102.989	4.009
2700	109.987	106.998	2.989
2800	108.968	97.011	11.957
3000	109.987	90.014	19.973
3250	87.025	87.025	0
3800	0	0	0

5DFL01ZA 100% row (MM3S)

RPM	A	B	A – B
600	100	100	0
800	67.527	67.527	0
1000	68.003	68.003	0
1200	88.315	84.987	3.328
1280	96.468	95.517	0.951
1400	94.633	94.022	0.611
1580	94.973	93.207	1.766
1800	95.177	93.818	1.359
2000	97.011	93.614	3.397
2200	98.03	93.275	4.755

2400	99.525	94.973	4.552
2500	100.68	95.517	5.163
2600	104.008	96.468	7.54
2700	107.473	100.272	7.201
2800	106.998	94.022	12.976
3000	109.987	87.296	22.691
3250	87.025	87.025	0
3800	0	0	0

FULL 4 × 18 GRIDS

Low-load rows match; only the 100% row moves

5DFL00ZA transient · 4 × 18 · ITN 1059 · amber = A ≠ B

Y \ X	600	800	1000	1200	1280	1400	1580	1800	2000	2200	2400	2500	2600	2700	2800	3000	3250	3800
0	8.968	4.008	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
25	33.22	28.261	23.234	23.234	23.234	23.234	23.234	23.234	23.234	23.234	23.234	23.234	23.234	23.234	23.234	23.234	23.234	0
50	57.473	52.514	47.487	47.487	47.487	47.487	47.487	47.487	47.487	47.487	47.487	47.487	47.487	47.487	47.487	47.487	47.487	0
100	100	68.275	68.003	88.315 B 85.53	97.487	95.517 B 94.973	94.022	95.992 B 94.701	98.506 B 95.788	100 B 97.011	101.495 B 98.981	103.397 B 100.815	106.998 B 102.989	109.987 B 106.998	108.968 B 97.011	109.987 B 90.014	87.025	0

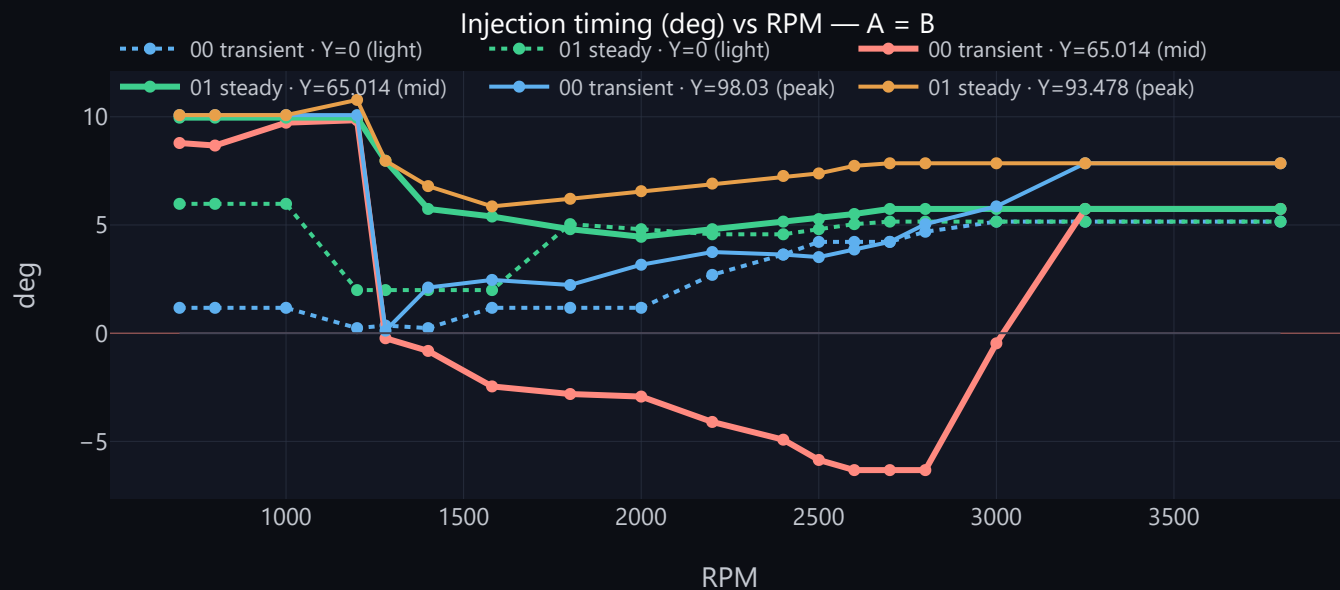
5DFL01ZA steady-state · 4 × 18 · ITN 1056 · amber = A ≠ B

Y \ X	600	800	1000	1200	1280	1400	1580	1800	2000	2200	2400	2500	2600	2700	2800	3000	3250	3800
0	8.968	4.008	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
25	33.017	27.989	23.03	23.03	23.03	23.03	23.03	23.03	23.03	23.03	23.03	23.03	23.03	23.03	23.03	23.03	23.03	0
50	56.998	51.97	47.011	47.011	47.011	47.011	47.011	47.011	47.011	47.011	47.011	47.011	47.011	47.011	47.011	47.011	47.011	0
100	100	67.527	68.003	88.315 B 84.987	96.468 B 95.517	94.633 B 94.022	94.973 B 93.207	95.177 B 93.818	97.011 B 93.614	98.03 B 93.275	99.525 B 94.973	100.68 B 95.517	104.008 B 96.468	107.473 B 100.272	106.998 B 94.022	109.987 B 87.296	87.025	0

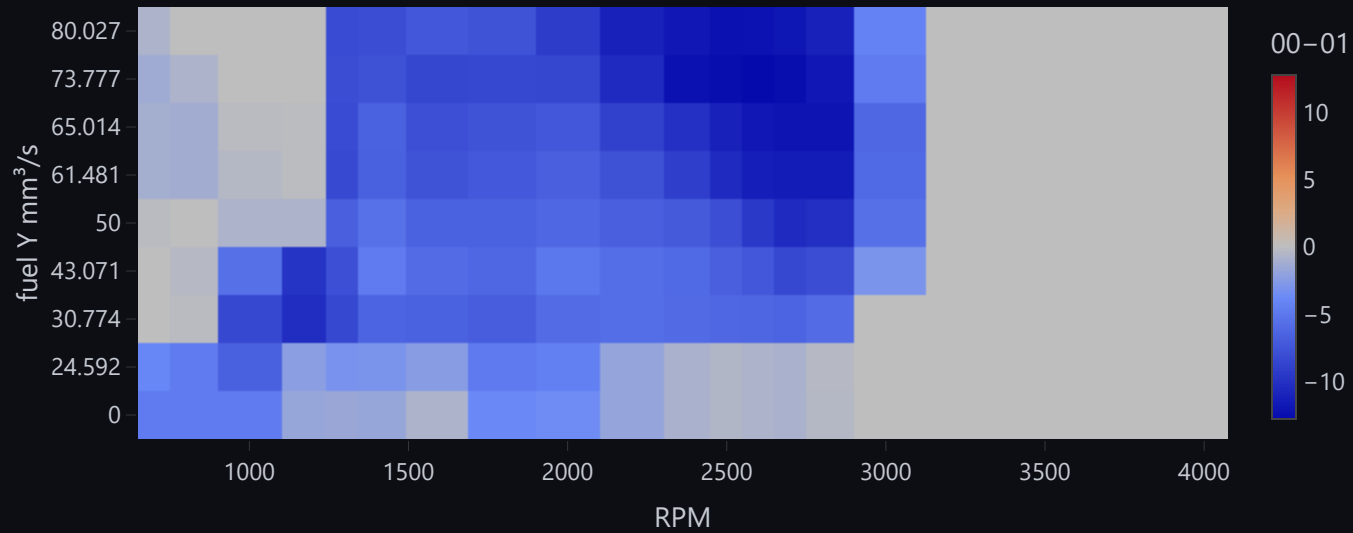
4DTA00ZA / 4DTA01ZA · A = B

Timing vs RPM — peak-load vs light, and the transient hole

Decoded 11×18 grids match A vs B. The contrast that matters is **transient (00)** vs **steady (01)**, not box vs box. Y is commanded fuel (mm³/s), not throttle percent. Peak Y is not shared: transient top 98.03 vs steady top 93.478.



Transient – steady (deg), shared Y

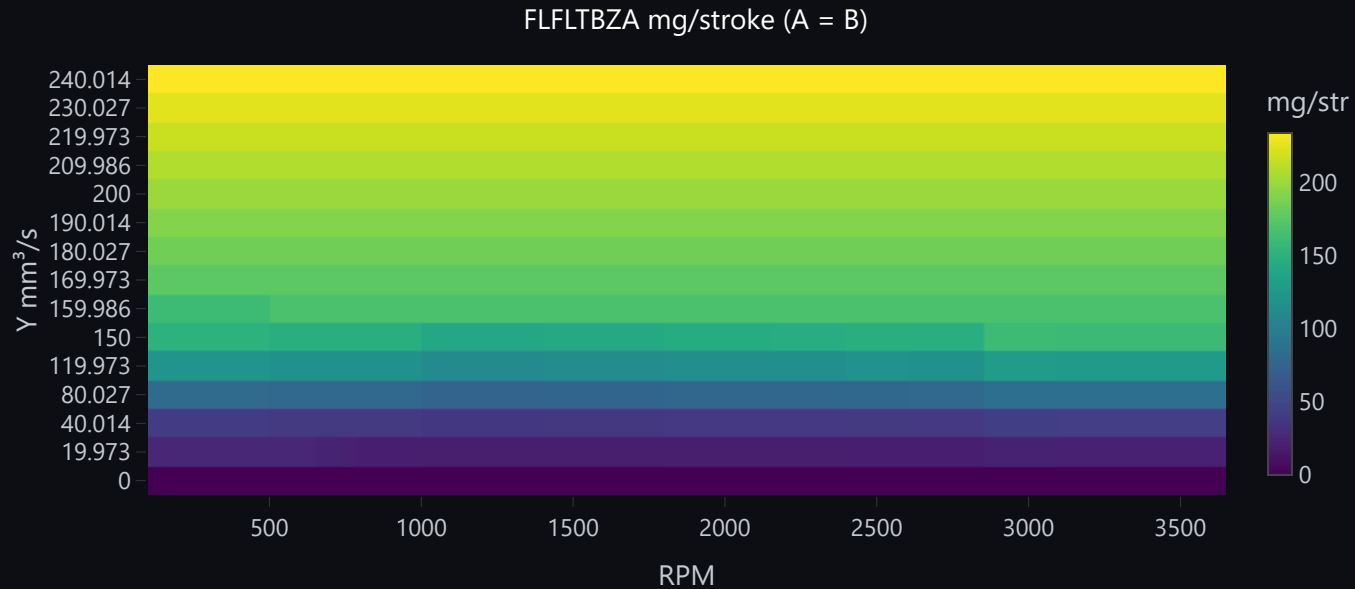


Deepest transient retard: **-6.328°** at 2700 RPM, Y=61.481 mm³/s on 4DTA00. From 1280 through ~2800 RPM the transient table goes negative at mid/high fuel (~-2 to -6.3°). Steady-state stays positive in that band (~+4 to +8°). At 3000+ both come back positive. 3800 repeats 3250 on most rows.

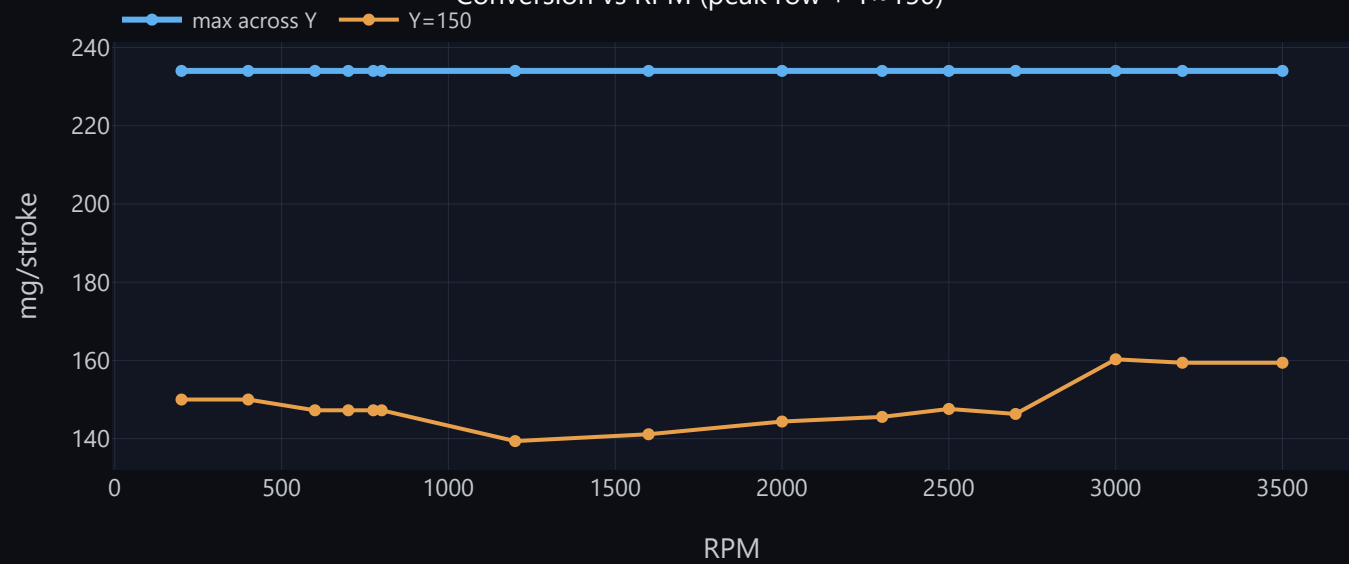
FLFLTBZA · A = B · 15 × 15 · ITN 8014

Fuel units → mg/stroke — conversion, not the request

This is the density-style conversion table. Changing it without changing 5DFL is a different edit than raising the 100% fuel row. Below $\sim 160 \text{ mm}^3/\text{s}$ the table still varies with RPM (dip 1200–2700, bump at 3000). From $Y \approx 160$ up, rows are flat across RPM.



Conversion vs RPM (peak row + $Y \approx 150$)



AFC limiter — compare by breakpoint, not column index

truncated Z dump 512/588 bytes (256/294 cells) — 256 of 294 cells present. Last complete row is Y index 11; Y index 12 has four cells; Y index 13 is empty. Do not treat the declared 14×21 as fully known. The 29-map HTML viewer is index-aligned and will mislead on this map.

RPM X AFFLLMXA

A: 600, 700, 800, 1000, 1200, 1280, 1400, 1580, 1800, 2000, 2200, 2400, 2500, 2600, 2700, 2800, 3000, 3250, 3800, 4000, 4200

B: 600, 800, 1000, 1200, 1280, 1400, 1580, 1800, 2000, 2200, 2400, 2500, 2600, 2700, 2800, 3000, 3250, 3800, 4000, 4200, 4500

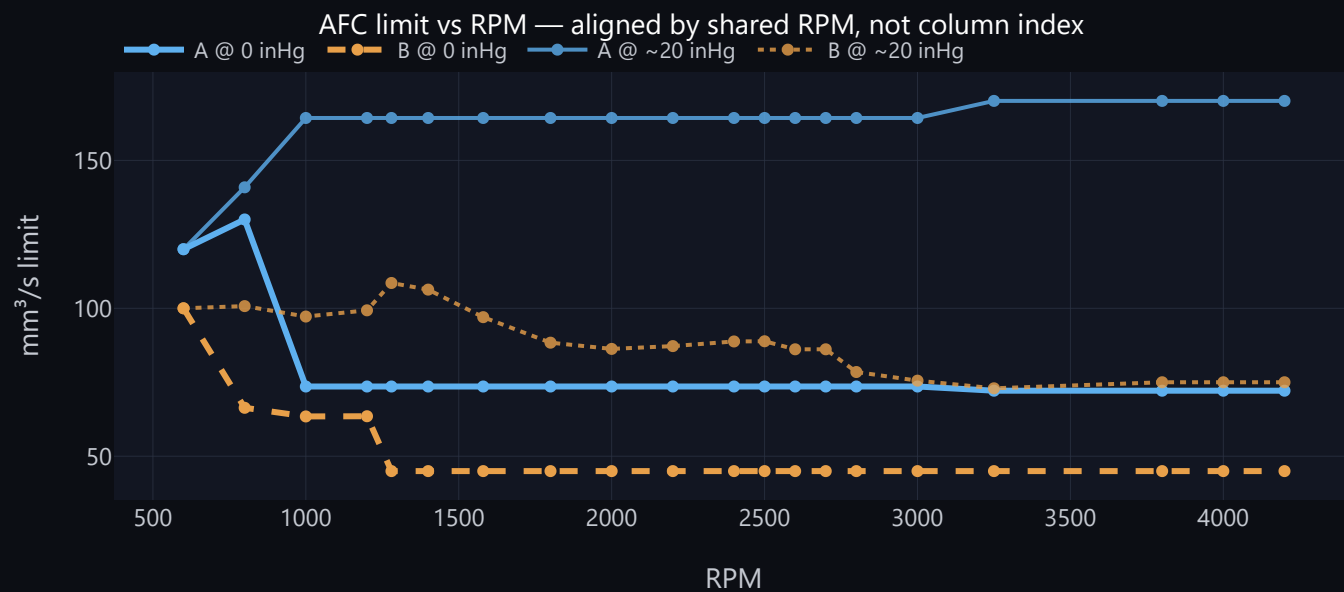
A only: 700 · B only: 4500

BOOST Y AFFLLMYA (INHG)

A: 0, 1.002, 2.004, 3.006, 4.995, 10.005, 15, 19.994, 25.005, 29.999, 34.994, 40.004, 43.997, 48.005

B: 0, 1.002, 2.004, 3.006, 4.995, 10.005, 15, 19.994, 34.994, 40.004, 43.997, 48.005, 100.003, 110.008

A only: 25.005, 29.999 · B only: 100.003, 110.008 (100/110 inHg is above stock Dodge boost — leftover tail).



AFFLLMZA at 0 inHg (zero boost), shared RPM only

	RPM	A	B	A - B
	600	119.973	100	19.973
	800	130.027	66.372	63.655
	1000	73.573	63.451	10.122
	1200	73.573	63.519	10.054
	1280	73.573	44.973	28.6
	1400	73.573	44.973	28.6
	1580	73.573	44.973	28.6
	1800	73.573	44.973	28.6
	2000	73.573	44.973	28.6
	2200	73.573	44.973	28.6
	2400	73.573	44.973	28.6
	2500	73.573	44.973	28.6
	2600	73.573	44.973	28.6
	2700	73.573	44.973	28.6
	2800	73.573	44.973	28.6
	3000	73.573	44.973	28.6

3250	72.147	44.973	27.174
3800	72.147	44.973	27.174
4000	72.147	44.973	27.174
4200	72.147	44.973	27.174

AFFLLMZA at 19.994 inHg (~10 psi), shared RPM only

RPM	A	B	A – B
600	119.973	100	19.973
800	140.897	100.747	40.15
1000	164.334	97.215	67.119
1200	164.334	99.321	65.013
1280	164.334	108.56	55.774
1400	164.334	106.318	58.016
1580	164.334	97.011	67.323
1800	164.334	88.383	75.951
2000	164.334	86.277	78.057
2200	164.334	87.228	77.106
2400	164.334	88.791	75.543
2500	164.334	88.859	75.475
2600	164.334	86.141	78.193
2700	164.334	86.141	78.193
2800	164.334	78.465	85.869
3000	164.334	75.543	88.791
3250	170.109	72.962	97.147
3800	170.109	75	95.109
4000	170.109	75	95.109
4200	170.109	75	95.109

At zero boost, B falls to a 44.973 mm³/s ceiling from 1280 RPM up; A holds 73.573 (then 72.147 at 3250+). At ~20 inHg A sits near 164 mm³/s across most RPM — above any 5DFL 100% cell — while B is lower and RPM-shaped. On the sampled complete rows, A's AFC ceiling is well above 5DFL; B's AFC is much tighter at low boost. This is a third decoded-table difference, incomplete.

What this means if you tune

A is the richer WOT fuel cal. Timing is identical on these two boxes, so *fueling* is the lever between them — not a hidden timing advance on J90269.06.

- Part-throttle 5DFL (0 / 25 / 50) matches. Cruise / light load is not the delta.
- The 100% row is the complete, trustworthy cal delta: A holds ~107–110 mm³/s at 2800–3000; B drops into the high-80s / 90s.
- Steady-state 5DFL01 is slightly leaner than transient 5DFL00 on both boxes, and the A-vs-B gap is *larger* there (+22.691 vs +19.973 at 3000).
- FLFLTBZA matches — you are not looking at a conversion-table difference.
- AFC also differs (A richer limiter at shared low-boost breakpoints) but the dump is truncated. Do not index-align it.
- A piggyback that already adds fuel stacked on A's richer 5DFL is more fuel than either change alone. This page does not tell you how to program an ECM.

THIS IS NOT

Not a flash / ROM image. Not a download-to-ECM file. Not a pump-frame recipe. Not overlay AID curves. Decoded values are engineering units after scale/add from a live upload.

Full grids: [tune A vs B.html](#) · [factory-fuel.md](#) · [factory-timing.md](#)

SOURCES

Where the numbers came from

Chart data is generated from `maps/tune_A_vs_B.json` (re-run `maps/build_factory_a_vs_b.py`). Identity from `A_identity.json` / `B_identity.json`. Cells are not invented.

Smarty handheld SW capture on a spare is **future work** (owner will flash level 6 then level 4 on that box). It is not in this dump and not the point of this deck.

Read-only documentation. Open as a file in a browser; charts load Plotly from a CDN (tables still work offline).

